evaporated. Had Earth been much farther away, the oceans would have frozen. In either case, life as we know it would not have evolved.

Within the chemically rich liquid oceans, by a mechanism yet to be discovered, organic molecules transitioned to self-replicating life. Dominant in this primordial soup were simple anaerobic bacteria—life that thrives in oxygen-empty environments but excretes chemically potent oxygen as one of its by-products. These early, single-celled organisms unwittingly transformed Earth's carbon dioxide-rich atmosphere into one with sufficient oxygen to allow aerobic organisms to emerge and dominate the oceans and land. These same oxygen atoms, normally found in pairs (O_2), also combined in threes to form ozone (O_3) in the upper atmosphere, which serves as a shield that protects Earth's surface from most of the Sun's molecule-hostile ultraviolet photons.

We owe the remarkable diversity of life on Earth, and we presume elsewhere in the universe, to the cosmic abundance of carbon and the countless number of simple and complex molecules that contain it. There's no doubt about it: more varieties of carbon-based molecules exist than all other kinds of molecules combined.

But life is fragile. Earth's occasional encounters with large, wayward comets and asteroids, a formerly common event, wreaks intermittent havoc upon our ecosystem. A mere sixty-five million years ago (less than two percent of Earth's past), a ten-trillion-ton asteroid hit what is now the Yucatan Peninsula and obliterated more than seventy percent of Earth's flora and fauna—including all the famous outsized dinosaurs. Extinction. This ecological catastrophe enabled our mammal ancestors to fill freshly vacant niches, rather than continue to serve as hors d'oeuvres for *T. rex*. One big-brained branch of these mammals, that which we call primates, evolved a genus and species (*Homo sapiens*) with sufficient intelligence to invent methods and tools of science—and to deduce the origin and evolution of the universe.

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What happened before all this? What happened before the beginning?

Astrophysicists have no idea. Or, rather, our most creative ideas have little or no grounding in experimental science. In response, some religious people assert, with a tinge of righteousness, that *something* must have started it all: a force greater than all others, a source from